Metadata Aware

Utilizing Metadata for Better Retrieval-Augmented Generation
Raquib Bin Yousuf et al. (ECIR 2026)

Paper: https://arxiv.org/abs/2601.11863

What the paper proposes

The paper studies whether titles, document types, sections, company fields, and other metadata can disambiguate repetitive chunks better than text-only similarity.

What changed in this project

Encoded the project metadata fields—title, source document, source type, document type, manufacturer, vehicle model, category, and group—and fused metadata similarity with content similarity at 8% metadata weight and 92% content weight.

Measured results

Metric	Baseline	Method	Change
Hit@1	80.0%	85.0%	+5.0 pp
Hit@5	85.0%	95.0%	+10.0 pp
Recall@5	64.7%	73.1%	+8.3 pp
NDCG@10	84.0%	86.0%	+2.0 pp
MRR@20	82.9%	87.7%	+4.8 pp
Latency	26.9 ms	26.3 ms	—

Inference

This was the strongest balanced improvement: Hit@1 rose to 85.0%, Hit@5 to 95.0%, Recall@5 to 73.1%, NDCG@10 to 86.0%, and MRR@20 to 87.7%. Latency was 26.3 ms.